

TASK 3 :
DCSE UET PESHAWAR
MBSD Spring 2024

- A. Generate a signal on pin P1.1 having frequency equal to 10 Hz with a duty cycle of 10%.**
- B. When a user presses a button at P1.2 then frequency changes to 20Hz with a 20% duty cycle.**
- C. When a user again presses the same button then frequency changes to 40Hz with a duty cycle of 40%.**
- D. When a user again presses the same button then frequency changes to 80Hz with a duty cycle of 80%.**
- E. Show it on an oscilloscope.**
- F. Each time a user presses a button the signal toggles from case A to B, then B to C, then C to D and finally from D to A, on every subsequent button press.**
- G. Program only in C.**
- H. Use Timers to generate delays. I. Button press means, press and release. Press doesn't mean you have to keep the button pressed without releasing it. J. Use only Proteus**